

ESP32 SECURITY SUITE

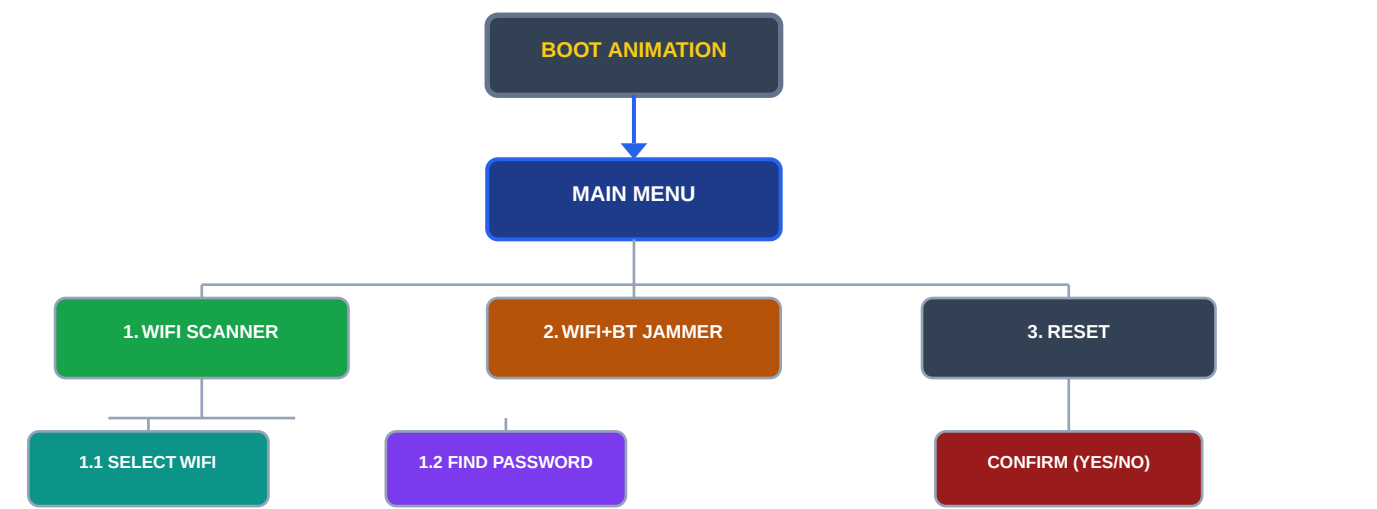
TFT Display UI Design Model — CradleGuard 3.0

2.4" MCUFRIEND TFT LCD | 320 × 240 px | Landscape

LEGAL DISCLAIMER:

This system is strictly for authorized security research and educational use in controlled environments.

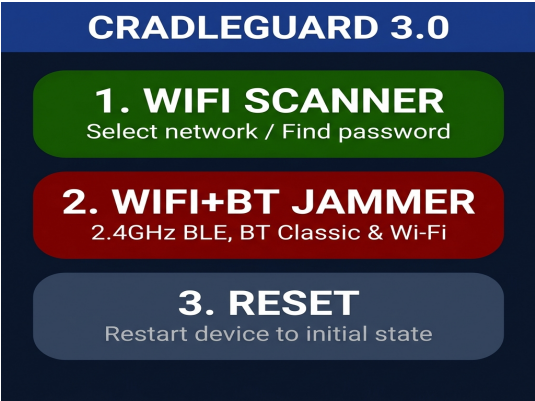
Screen Navigation Flow



Navigation Controls: BOOT button (GPIO 0) = GO BACK one level | Tap header "< ..." = GO BACK | Physical ON btn = context action

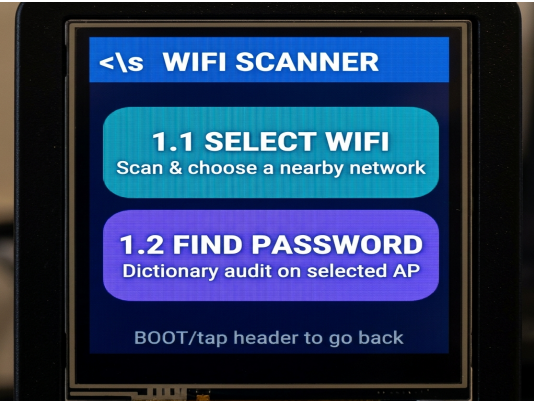
Screen Designs — Rendered Mockups

Screen 1: Main Menu



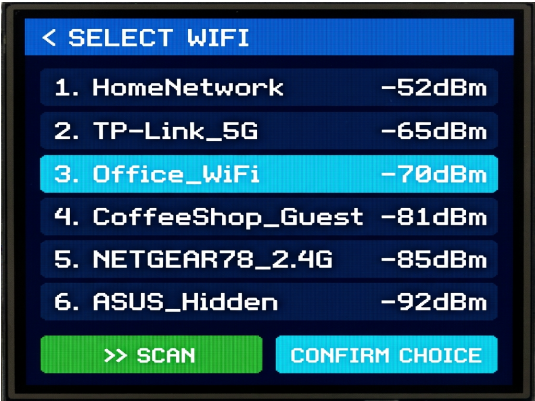
- Header: "CRADLEGUARD 3.0" (Blue bar)
- Button 1: WIFI SCANNER !' Dark Green !' GPIO 12 tap or touch
- Button 2: WIFI+BT JAMMER !' Dark Red !' Activates dual NRF24
- Button 3: RESET !' Slate Blue !' Calls ESP.restart()

Screen 2: Wi-Fi Scanner Sub-Menu



- "< WIFI SCANNER" header !' tap to go back
- Button 1.1: SELECT WIFI !' Teal !' Scans & lists APs
- Button 1.2: FIND PASSWORD !' Violet !' Runs wordlist audit
- BOOT button (GPIO 0) returns to Main Menu

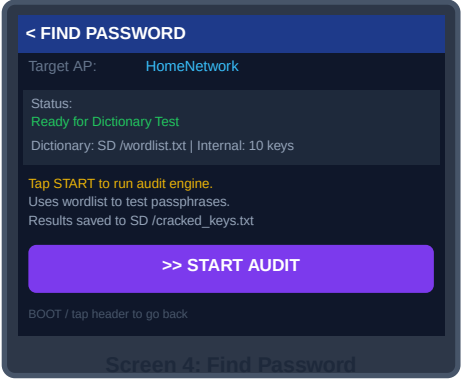
Screen 3: 1.1 Select Wi-Fi Network



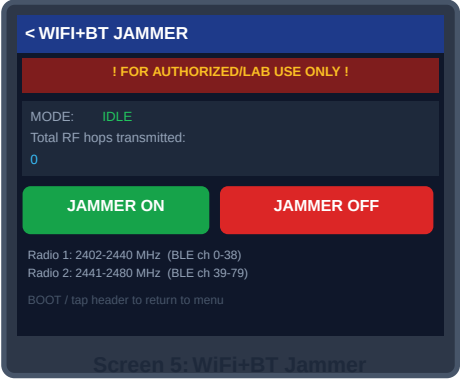
- Header "< SELECT WIFI" !' tap to return to sub-menu
- List rows: tap any row to highlight/select that network
- Highlighted row !' filled Cyan, text dark
- SCAN button !' Green, triggers scanWiFiNetworks()
- CONFIRM CHOICE !' Cyan, saves selectedTargetIndex & goes back
- Physical TURN ON (GPIO 12) also triggers a new scan

Screen Designs — Diagram Mockups

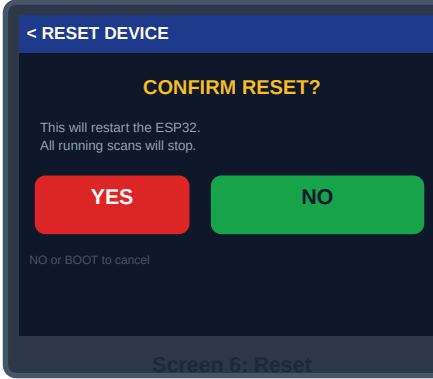
Screen 4: 1.2 Find Password / Dictionary Audit



- Tap START AUDIT !' calls runDictionaryCrack()
- Result shows in Status box (green = found)
- cracked_keys.txt written to SD card



- JAMMER ON !' actionActive = true, LED on
- JAMMER OFF !' actionActive = false
- RF hops counter updates live every 500ms



- YES !' ESP.restart() after 1.5s delay
- NO / BOOT !' back to Main Menu
- Displays "RESTARTING..." before reset

Touch Zone Map & Control Reference

Screen	Touch Zone / Action	ESP32 Event Triggered	GPIO / Button
Main Menu	Tap WIFI SCANNER row	setScreen(SCREEN_WIFI_SUBMENU)	Touch
Main Menu	Tap WIFI+BT JAMMER row	setMode(BLE) + setScreen(SCREEN_JAMMER)	Touch
Main Menu	Tap RESET row	setScreen(SCREEN_RESET)	Touch
WiFi Sub	Tap 1.1 SELECT WIFI	setScreen(SCREEN_WIFI_SELECT) + scan	Touch
WiFi Sub	Tap 1.2 FIND PASSWORD	setMode(WIFI_CRACK) + crack screen	Touch
WiFi Sub	Tap header "<"	setScreen(SCREEN_MAIN_MENU)	Touch / BOOT btn
Select WiFi	Tap a network row	selectedTargetIndex = i; redraw	Touch
Select WiFi	Tap >> SCAN	scanWiFiNetworks(); redraw list	Touch / GPIO 12
Select WiFi	Tap CONFIRM CHOICE	setScreen(SCREEN_WIFI_SUBMENU)	Touch
Find Password	Tap >> START AUDIT	runDictionaryCrack(); redraw	Touch / GPIO 12
Find Password	Tap header "<"	setScreen(SCREEN_WIFI_SUBMENU)	Touch / BOOT btn
Jammer	Tap JAMMER ON	actionActive = true; LED HIGH	Touch / GPIO 12
Jammer	Tap JAMMER OFF	actionActive = false; LED LOW	Touch / GPIO 14
Jammer	Tap header "<"	setScreen(SCREEN_MAIN_MENU)	Touch / BOOT btn
Reset	Tap YES	delay(1500) + ESP.restart()	Touch
Reset	Tap NO	setScreen(SCREEN_MAIN_MENU)	Touch / BOOT btn

Physical Button Mapping

Button	GPIO Pin	Wiring	Context Action
TURN ON Button	GPIO 12	One leg !' GPIO 12, other !' GND (INPUT_PULLUP)	Scan / Start Audit / Jammer ON
TURN OFF Button	GPIO 14	One leg !' GPIO 14, other !' GND (INPUT_PULLUP)	Jammer OFF / Stop Action
BOOT Button	GPIO 0	Built-in ESP32 BOOT pushbutton	Navigate BACK one screen level
Status LED	GPIO 2	Built-in ESP32 LED	HIGH when jammer/action is active

UI Color Palette Reference

Deep Navy (BG) Screen background	Header Blue Top header bars	Dark Green WIFI SCANNER button / JAMMER ON
Dark Red WIFI+BT JAMMER button / JAMMER OFF	Teal SELECT WIFI button	Violet FIND PASSWORD / START AUDIT button
Slate RESET button background	Cyan Accent Selected items / CONFIRM button	Yellow Warning Warning bars and instruction text